

Random and mixed effects models

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LSM

Ex 1. A small company has 5 personnel officers, each one of whom interviews several candidates for a job, and assigns a score for each candidate.

It is of interest to know if there is any difference variation in ratings of candidates by different officers
→ one-factor ANOVA model is applicable

Ex 2. A big company has many personnel officers. 5 of them are randomly selected, and these selected officers each interview several candidates for a job and assign scores

- (a) Is there any variation in ratings across different officers in the company?
- (b) How to estimate the mean score?

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij} \quad \begin{matrix} j=1, \dots, n_i \\ i=1, \dots, r \end{matrix}$$

r = # of officers selected

n_i = # candidates interviewed by the i^{th} (selected) officers

Assumptions: ① τ_i 's are independent with zero mean and variance σ_μ^2 [Stronger: $\tau_i \stackrel{\text{iid}}{\sim} N(0, \sigma_\mu^2)$]

② ϵ_{ij} 's are iid with mean zero and variance σ^2

③ τ_i 's and ϵ_{ij} 's are indep distributed.

Distribution of the observation:

$$E Y_{ij} = \mu \quad \text{Var} Y_{ij} = \sigma_{\mu}^2 + \sigma^2$$

(Under Gaussianity, $Y_{ij} \sim N(0, \sigma_{\mu}^2 + \sigma^2)$)

Also if $i \neq i'$ Y_{ij} and $Y_{i'j'}$ are indept.

$$\begin{aligned} \text{Cov}(Y_{ij}, Y_{i'j'}) &= \text{Cov}(\tau_i + \epsilon_{ij}, \tau_{i'} + \epsilon_{i'j'}) \\ &= \text{Var} \tau_i = \sigma_{\mu}^2 \end{aligned}$$

(a) Whether there is any variation in average ratings by the officers in the company?

$$H_0: \sigma_{\mu}^2 = 0 \quad \text{vs.} \quad H_A: \sigma_{\mu}^2 > 0$$

(b) Estimation of overall mean rating μ

Inference questions.

(a) $H_0: \sigma_{\mu}^2 = 0$ vs. $H_A: \sigma_{\mu}^2 > 0$ (variation in treatment effects)

"Raw" estimate of τ_i is $\bar{Y}_{i.} - \bar{Y}_{..}$ (from one way ANOVA model)

"Raw" estimate of μ is $\bar{Y}_{..}$ (unbiased)

ANOVA decomposition

$$\tau_i = \mu_i - \mu$$

$$SST_0 = SST_R + SSE$$

$$SST_R = \sum_{i=1}^r n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2$$

Distribution of quantities

$$SSE = \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2 \sim \sigma^2 \chi_{n_T - r}^2$$

Check that

$$E(MSTR) = \sigma^2 + N\sigma_\mu^2 \text{ where } N = \frac{1}{r-1} \left[\sum_{i=1}^r n_i - \frac{\sum n_i^2}{\sum n_i} \right]$$

$$df(SSR) = r-1$$

if $n_i = n$ for all i (balanced design) then $N = n$

$$F = \frac{MSTR}{MSE} \quad \text{Reject } H_0: \sigma_\mu^2 = 0 \text{ for large values of } F$$

Under balanced design i.e. $n_1 = \dots = n_r = n$ we can check that

$$\frac{SSR}{\sigma^2 + n\sigma_\mu^2} \sim \chi_{r-1}^2 \text{ (Exercise)}$$

$$\text{Under a balanced design: } F \stackrel{d}{=} \left(\frac{\sigma^2 + n\sigma_\mu^2}{\sigma^2} \right) F_{r-1, n_T-r}$$

Test: reject H_0 at level α if $F > F_{r-1, n_T-r}^{1-\alpha}$

Illustration: For the big company job interview problem,

Balanced design $n=4$
 $r=5$

officer	$i=$	Applicant			
		$j=1$	2	3	4
A	1	76	65	85	74
B	2	59	75	81	67
C	3	49	63	61	46
D	4	74	71	85	89
E	5	66	84	80	79

$$MSTR = 394.9$$

$$MSE = 73.3$$

$$F = \frac{MSTR}{MSE} = 5.39$$

$$F_{4,15}^{0.95} = 3.06$$

Remark: If we want a confidence interval for σ_μ^2
 We can try to invert the test procedure
 In fact this approach can be used to obtain
 C.I. for $\frac{\sigma_\mu^2}{\sigma^2}$ (and the C.I. is one sided)

Focus on balanced design only

Suppose, we want to obtain C.I. for μ

Point estimate: $\hat{\mu} = \bar{Y}_{..}$ (unbiased)

$$\text{Var } \hat{\mu} = \text{Var}\left(\frac{1}{r} \sum_{i=1}^r \bar{Y}_{i.}\right) = \frac{1}{r} \text{Var } \bar{Y}_{i.}$$

$$\text{Var } \bar{Y}_{i.} = \frac{1}{n^2} \text{Var}\left(\sum_{j=1}^n Y_{ij}\right)$$

$$= \frac{1}{n^2} \left\{ \sum_{j=1}^n \text{Var } Y_{ij} + \sum_{j \neq j'} \text{Cov}(Y_{ij}, Y_{ij'}) \right\}$$

$$= \frac{1}{n^2} \left\{ n(\sigma_\mu^2 + \sigma^2) + n(n-1)\sigma_\mu^2 \right\}$$

$$= \frac{1}{n} \left\{ \sigma_\mu^2 + \sigma^2 + (n-1)\sigma_\mu^2 \right\} = \frac{\sigma^2 + n\sigma_\mu^2}{n}$$

$$\therefore \text{Var } \hat{\mu} = \frac{\sigma^2 + n\sigma_\mu^2}{nr}$$

An unbiased estimate of $\text{Var } \bar{Y}_{..}$ is $s^2(\bar{Y}_{..}) = \frac{\text{MSTR}}{nr}$

Check that $\bar{Y}_{..}$ and MSTR are independent

Hint: $\bar{Y}_{i.} - \bar{Y}_{..} \perp \bar{Y}_{..}$

$$\frac{SSTR}{\sigma^2 + n\sigma_{\mu}^2} \sim \chi^2_{r-1}$$

$$\text{So, } \frac{\bar{Y}_{..} - \mu}{s(\bar{Y}_{..})} \sim t_{r-1}$$

$$\Rightarrow 100(1-\alpha)\% \text{ CI for } \mu \text{ is } \bar{Y}_{..} \pm t_{r-1}^{1-\alpha/2} \sqrt{\frac{MSTR}{nr}}$$

(c) Inference on intraclass correlation coefficient

$$\rho = \text{Corr}(Y_{ij}, Y_{ij'}) \text{ for } j = j'$$

$$= \frac{\sigma_{\mu}^2}{\sigma^2 + \sigma_{\mu}^2} = \text{proportion of variability in the measurements attributed to variability in the treatment effect}$$

$\sigma^2_{I_n} + \sigma^2_{\mu} \mathbf{1}_n \mathbf{1}_n^T \leftarrow$

$\text{Cov}(\underline{Y}) =$

	0	0
0		0
0	0	

Obtain confidence interval for p

Note that $\rho = \frac{\sigma_\mu^2}{\sigma^2 + \sigma_\mu^2} = \frac{\sigma_\mu^2/\sigma^2}{1 + \sigma_\mu^2/\sigma^2}$

Also recall $\frac{MSTR}{MSE} \sim \frac{\sigma^2 + n \frac{\sigma_\mu^2}{\sigma^2}}{\sigma^2} F_{r-1, n_T-r}$